Baber Rehman

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Summary

AI compiler engineer at Huawei Hong Kong Research Center, building compilers for Ascend NPUs: TileLang-Ascend, Triton-Ascend, AscendNPU-IR, and Triton-distributed-ascend. Holding a PhD in programming languages from the University of Hong Kong. Program committee service includes ECOOP, OOPSLA, POPL, and PLDI.

Academics

2018 - 2023 PhD in Computer Science at **University of Hong Kong**, Hong Kong
Research Area: Programming Languages, Type Systems
Functional Programming, Object-Oriented Programming
Supervisor: Dr. Bruno C. d. S. Oliveira

2011 - 2015 Bachelor's Degree at **University of Punjab**, Lahore, Pakistan (CGPA: 3.81/4.0)
Major: Computer Science
Result: Distinction (Class: 1/54, Batch: 2/191)

Professional Experience

Industry Experience

AI Compiler Engineer Nov 2023 - Present

Huawei, Hong Kong Research Center

- AI compiler engineer for Huawei Ascend NPUs. Working on TileLang-Ascend, Triton-Ascend, AscendNPU-IR, and Triton-distributed-ascend: the stacks that take tile-level kernels through MLIR and lower them onto the NPU.
- On TileLang-Ascend I work on the AscendNPU-IR path in the open compiler, including MLIR dialects and verifiers, JIT compilation, Ascend A5 operator lowering, and the tests and CI for that pipeline. On Triton-Ascend I work on the Ascend compiler backend, including static checks on kernel scope. Triton-distributed-ascend extends the same stack to distributed kernels. Related public work is a Lean model of Triton and the Triton-Ascend core, and a Lean core for Triton-Distributed.
- Earlier, created a P4-inspired DSL for programmable networking, a high-level way to describe packet-processing logic that compiles to data-plane code.

Software Engineer (DevOps) – Team Lead June 2015 - Aug 2018

CloudPlex PVT LTD, Lahore, Pakistan

- Led a team in Lahore and a remote team in Argentina, including backend hiring and client requirements, demos, and delivery.
- Built a cloud platform that automated orchestration and deployment on AWS and Azure (VPC, S3, EC2, VNet), using Golang, Chef, Ansible, Docker, and shell scripting.
- Automated firewall configuration for PfSense and Fortinet.

Intern Jul 2014 - Sep 2014

Mentor Graphics, Lahore, Pakistan

- Debugged Sourcery CodeBench, an Eclipse-based IDE, and fixed GUI bugs in Java and SWT.

Academic Experience

Resident Tutor Sep 2019 - Dec 2023

New College, The University of Hong Kong

– Supported students academically and socially, with mental health first aid and suicide prevention.
– Responsible for the Photography, Web, and News and Media teams.

Teacher Assistant Sep 2018 - Jun 2019

The University of Hong Kong

– Functional Programming, Principal of Programming Languages

Visiting lecturer Oct 2015 - Jun 2016

University of the Punjab, Lahore, Pakistan

– Fall 2015: Computer organization and Assembly Language course
– Spring 2016: Enterprise Application Development course (C#/MVC4)

Teacher Assistant Oct 2013 - Jul 2015

University of the Punjab, Lahore, Pakistan

– Discrete Mathematics, Data Structures and Algorithms, Database Systems, System Programming, Enterprise Application Development

Research Assistant Jun 2013 - Oct 2013

AIM Lab, University of the Punjab, Lahore, Pakistan

– Worked with business process management research repository

Technical Skills

Compilers	MLIR, LLVM, Triton, TileLang, AscendNPU-IR
Languages	C++, Python, Lean, Coq, Haskell, Scala, Golang
Cloud	AWS, Azure, Docker, Chef, Ansible, Git

Program Committees

– 2027: FormaliSE (AEC)
– 2026: ECOOP (PC), OOPSLA (AEC), POPL (AEC)
– 2025: ECOOP (AEC), OOPSLA (PC), PLDI (AEC)
– 2024: ECOOP (AEC, ERC), POPL (AEC), PLDI (AEC), ISSTA (AEC)
– 2023: ECOOP (AEC, ERC), POPL (AEC), ATC (AEC), OSDI (AEC)
– 2022: ECOOP (AEC, ERC), APLAS (AEC)

Selected Publications

– Disjoint Polymorphism with Intersection and Union Types. FTfJP 2024.
– A Blend of Intersection Types and Union Types. PhD dissertation, 2023.
– Correctness-by-Construction meets Refinement Types. FTfJP 2023.
– Union Types with Disjoint Switches. ECOOP 2022.
– The Duality of Subtyping. ECOOP 2020.